

FALOCHRON

Maritime Broadcast Fusion Node — the sea still broadcasts when the cloud and GNSS go dark

A portable, fully **offline** box that turns what the sea broadcasts — **NAVTEX** (text & 518 kHz radio) and **VHF voice** — into verified, structured, geo-referenced events (IMO S-124), correlates them in time and space, and puts them on a local map. **No cloud. No GNSS. Sub-billion models on-premise.**

~450 km

reach of GPS jamming/spoofing from Kaliningrad — waters of seven Baltic states. AIS & GNSS are the first casualties.

Situational awareness on the Baltic leans on AIS and GNSS — both actively degraded. Meanwhile **independent channels keep transmitting in the clear**: NAVTEX warnings, navigational-warning text, VHF reports. Today a **human** reads them. Nobody offers a small local device that fuses them into one auditable event picture and keeps working when the link — and the satellites — disappear. **We build the layer one step before every existing system**: ingestion & normalization that can feed GIS / TAK / trackers instead of competing with them.

BROADCAST IN

NAVTEX text (BHMW / NAVAREA)
SDR 518 kHz → SITOR-B
VHF voice → ASR

LOCAL CORE

SLM extracts to a strict schema (grammar-constrained)
→ **deterministic validator**
→ geo + time **correlator**

OUT

Offline map (MapLibre)
GeoJSON / CAP export
store-and-forward to a 2nd node

The SLM only proposes structure — it never decides, publishes or classifies intent. Form comes from the grammar; truth from validators, evidence and a human flag.

Civilian, every day

A marina / port operator gets an automatic board of live warnings & area closures on a map — no more reading NAVTEX by hand.

Dual-use, in a crisis

A coastal-resilience node that keeps a local event picture alive under GNSS jamming and a severed WAN — the demo literally unplugs the network mid-run.

Solo today — I want a crew of 2-4 for the 48 h.

Frontend / geo

MapLibre offline map, event cards, two-screen ops UI (no chat UI).

Backend

Go/Rust/Python service, SQLite event store, GeoJSON/CAP REST, store-and-forward.

ML / SLM

prompt + constrained decoding, the model cascade, QLoRA fine-tune on Modal.

RF / SDR & ASR

fldigi/SITOR-B, whisper.cpp on VHF, maritime domain / GIS know-how.

Come as you are — one strong dimension is enough. The prep (schema, benchmark, decoder, data, cache) is

WHY JOIN THIS ONE — THE HOMEWORK IS ALREADY DONE

Ready before the gun: a real pre-work base, not a slide idea

Everything below was built as FROM-SCRATCH-legal preparation (research, data, component tests — no pipeline code). It means the team spends the 48 h on the product, not on plumbing, and the pitch quotes measured numbers, not hopes.

✓ **Event schema, frozen**

S-124-aligned JSON Schema v0.1 + **GBNF grammar** for 100 %-valid constrained output; 10 labelled fixtures; grammar verified in real llama.cpp.

✓ **SLM benchmark harness**

9 models swept on CPU (0.35B → 4B) × 2 prompts: schema validity, field F1, **unsupported-fact rate** (safety), latency, RSS, Pareto — thresholds frozen before results.

✓ **Model cache, offline**

12 models / 14 GB (Granite, LFM2.5, Qwen3.5, Ministral, Phi-4-mini, Qwen3 + whisper small/med/large-v3-turbo) — sha256-manifested, USB-mirrored.

✓ **SITOR-B / NAVTEX decoder**

Headless fldigi rig proven on real off-air audio (**0 % CER** on a 2018 capture) + a bit-exact CCIR-476 signal generator for the demo's RF-in-a-box.

✓ **Cloud data factory**

Modal jobs (L4/A100): synthetic NAVTEX/VHF generation + LLM-judge, TTS augmentation, parallel bench, QLoRA — **train anywhere, run everywhere**; zero cloud in the demo.

✓ **Baltic data**

4 333-name gazetteer (names → coords, ODbL), 46 attributed real seeds + an MARTTS EN eval set, off-air 518 kHz recordings.

The 48-hour cut (scope shrinks text → map → VHF → SDR-live)

- 0–6 h** **Contract & skeleton** — schema v1, event store, llama.cpp server, contract tests.
- 6–14 h** **Text channel** — header parser + constrained extraction + validator; QLoRA starts on Modal in the background.
- 14–20 h** **Map + export** — events on the offline map; GeoJSON REST; event card with raw source.
- 20–36 h** **SDR & VHF** — IQ → fldigi → event; VAD → ASR → event; the geo/time correlator.
- 36–48 h** **Benchmark, kill-WAN, freeze** — the "smallest model that passes" table; two-node sync; pitch ×3.

What we are NOT building: a threat detector, a C2 platform, or another AIS tracker. We relay what was broadcast, with provenance — no intent classification, no autonomous action.

Why it wins with a jury: a real gap (broadcast-channel fusion exists nowhere), edge resilience you can demo by pulling the plug, open standards (S-124 / CAP), and a benchmark as the research artifact.